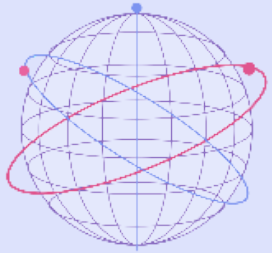
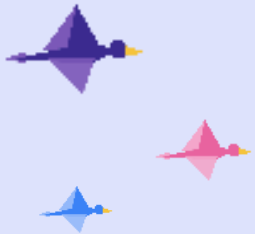
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
HEATSHIELD-Q: Quantum Allocation of Campus Cooling Resources

TRACK
Track 6: Climate & Sustainability

TEAM NAME
NEXORA

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INSTITUTION / ORGANISATION
KLE TECHNOLOGICAL UNIVERITY, Belagavi Campus

COUNTRY
India

Problem Statement

The challenge

- Climate change is increasing the risk of extreme heat events on campuses.
- Heat-relief resources (cooling units, shaded or air-conditioned spaces) are limited, so not every zone can be protected.
- Zones differ in population, heat-risk level and the resources they need, so the best allocation is not obvious.

Our aim

- HEATSHIELD-Q: a Qiskit-based quantum optimization system that allocates limited cooling resources across campus zones.
- It uses QAOA to search for allocations and compares them with classical optimization.

INPUTS
Population, heat-risk level and cooling units needed per zone

CONSTRAINT
A fixed budget of cooling units

GOAL
Maximise heat-risk exposure covered (population × risk)

Objectives & Novelty

Objectives

- Model campus cooling allocation as a binary optimization problem, one qubit per zone
- Build a QAOA solver in Qiskit that finds high-quality allocations under a unit budget
- Benchmark QAOA against exact search and a greedy method
- Test how results change from 4 to 12 zones

Novelty

- Applies QAOA to campus heat-risk planning, a sustainability problem rarely tackled with quantum methods
- Budget handled by an unbalanced penalty term, so no extra slack qubits are needed
- Penalty weights are tuned automatically so the lowest-energy state is the true best allocation
- Honest benchmarking: we report where QAOA helps and where it does not

Solution & Quantum Advantage

Campus zones (population, risk, units) → **QUBO with** budget penalty → **Ising Hamiltonian** (6 qubits) → **QAOA circuit** (Qiskit) → **Best allocation** (top measured)

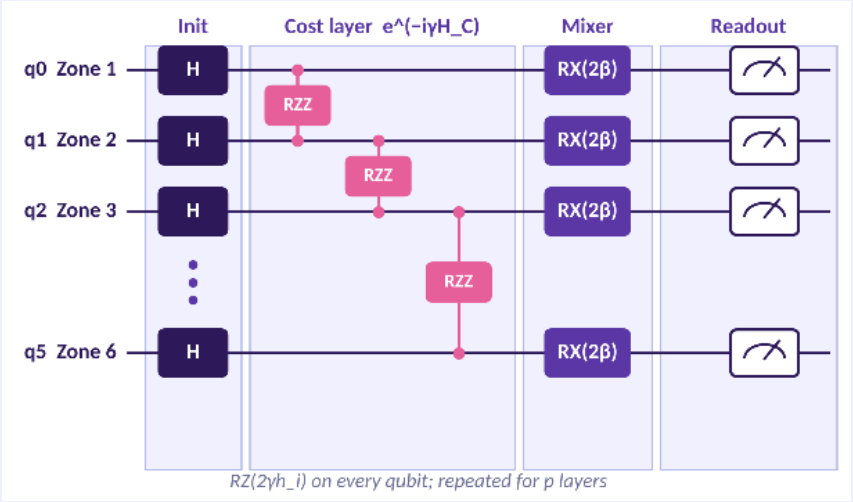
How it works

- Each qubit says whether a zone is funded (1) or not (0).
- The cost rewards covered exposure and penalises going over the unit budget.
- QAOA samples candidate allocations; the best-measured one is the answer.

Quantum advantage: our honest view

- At 6 to 12 zones, brute force is instant, so we claim no speed-up.
- QAOA finds the optimum more often than random guessing (up to 3.5× per shot).
- The method is a full pipeline that can be scaled to larger campuses and real hardware.

Algorithm & Circuit Design



Schematic: RZZ acts on every zone pair that has a cost term; shown for 4 of 6 qubits

Key details

- Qubits: 6 (one per zone)
- Start: Hadamard layer gives an equal superposition of all allocations
- Cost layer: RZ and RZZ gates from the Ising Hamiltonian (QUBO + unbalanced budget penalty)
- Mixer: RX rotations
- Parameters γ and β are tuned by a classical optimizer; depth p layers (1 to 4 explored)
- Measured with shots; the best-seen allocation is returned

Implementation & Tech Stack

1 Qiskit

QAOA circuit, simulation and sampling

2 heatshield/ package

Data, problem, classical baselines, Hamiltonian and QAOA modules

3 Streamlit app

Interactive demo; every number comes from Qiskit

4 run_hardware.py

Noisy IBM-chip simulator run; ready for real IBM hardware with an API key

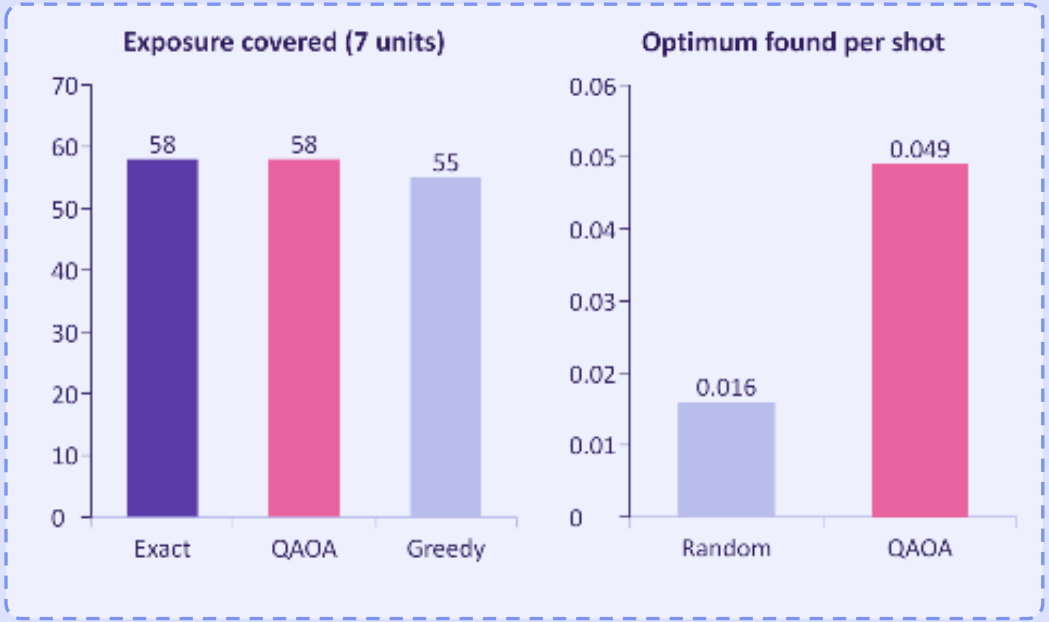
5 run_scaling.py

Scaling study from 4 to 12 zones

6 Tests + web view

5 unit tests (pass); browser visualisation in web/

Results & Analysis



Key details

- QAOA matches the exact optimum (58.0) and beats greedy (55.0) on the sample campus
- Optimum measured in 4.9% of shots vs 1.6% for random guessing
- 1,024 shots are enough to find it
- Scaling, 4 to 12 zones: 1.3 to 3.5× better than random; greedy stays within 2% of optimal
- Noisy IBM-chip simulation also run

Impact & Real-World Applications

1 Safer campuses

Directs scarce cooling resources to the zones where people and heat risk are highest.

2 Better planning

Gives administrators a repeatable, data-driven way to compare allocation options before a heatwave.

3 Climate resilience

Supports sustainability goals by using limited resources more effectively as heat events increase.

Where it applies: university campuses, schools, hospitals and city wards with limited heat-relief resources. The same model extends to any "fund the highest-impact zones within a budget" problem.

Challenges & Future Scope

Challenges

- Early QAOA runs found the optimum only about 1% of the time; tuning the penalty weights fixed this
- No quantum speed-up at 6 to 12 zones, because classical search is instant
- Results come from noise-free simulation; hardware noise will reduce quality
- Sample zone data is illustrative, not measured

Future scope

- Run on real IBM Quantum hardware and apply error mitigation
- Use real campus data: zone populations, shade and temperature readings
- Scale to larger zone counts and compare more classical solvers
- Add time-varying heat forecasts and multiple resource types

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-nexora-0022

DEMO VIDEO LINK
https://youtu.be/7Qv1oBpaJqU

